StudyPilot

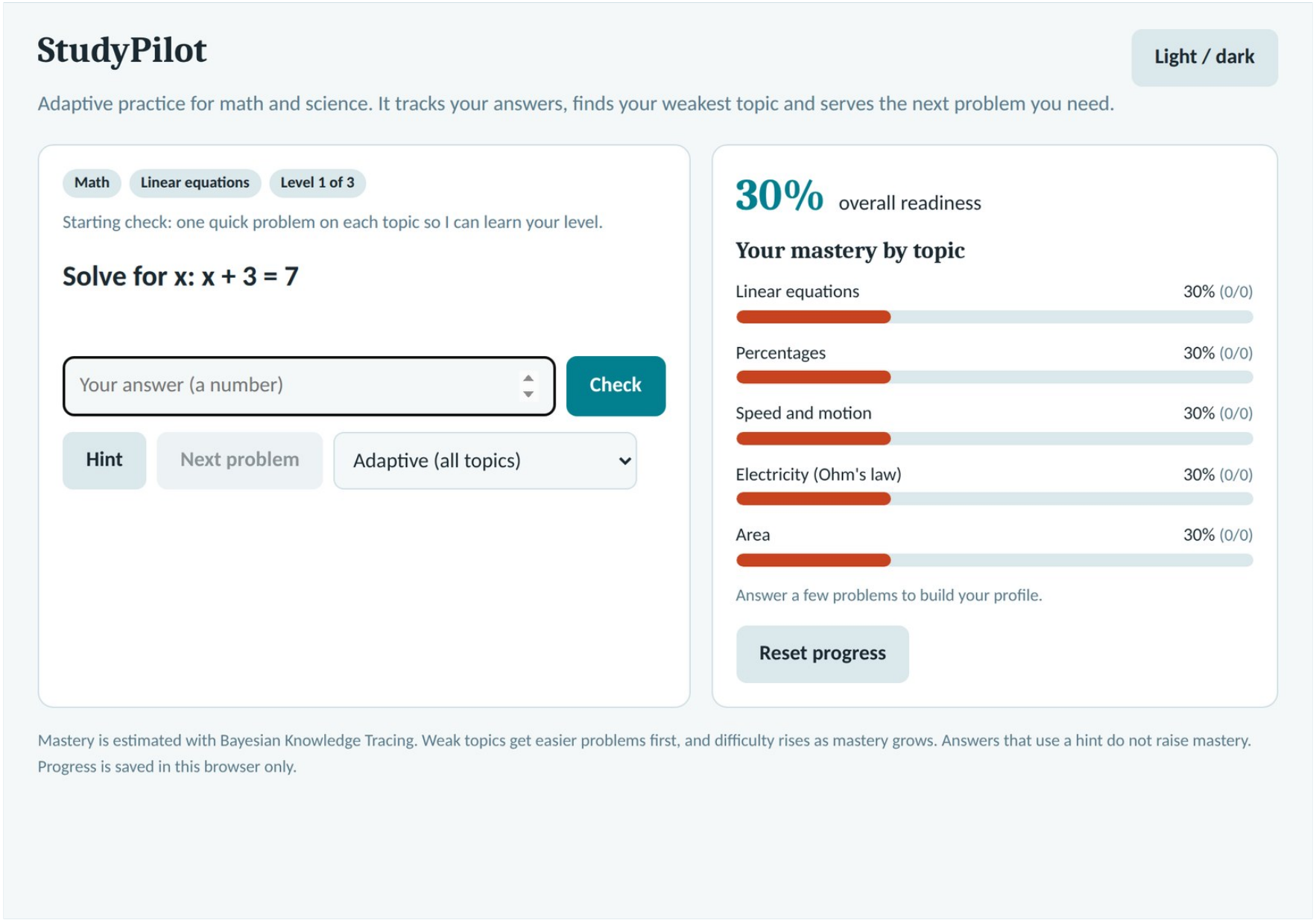
Adaptive practice for math and science

Project screenshots | IBM SkillsBuild Masterclass 5: Creating Real Life Projects

The app tracks a student's quiz answers, estimates mastery for each topic with Bayesian Knowledge Tracing, finds the weakest topics and serves customised practice problems.

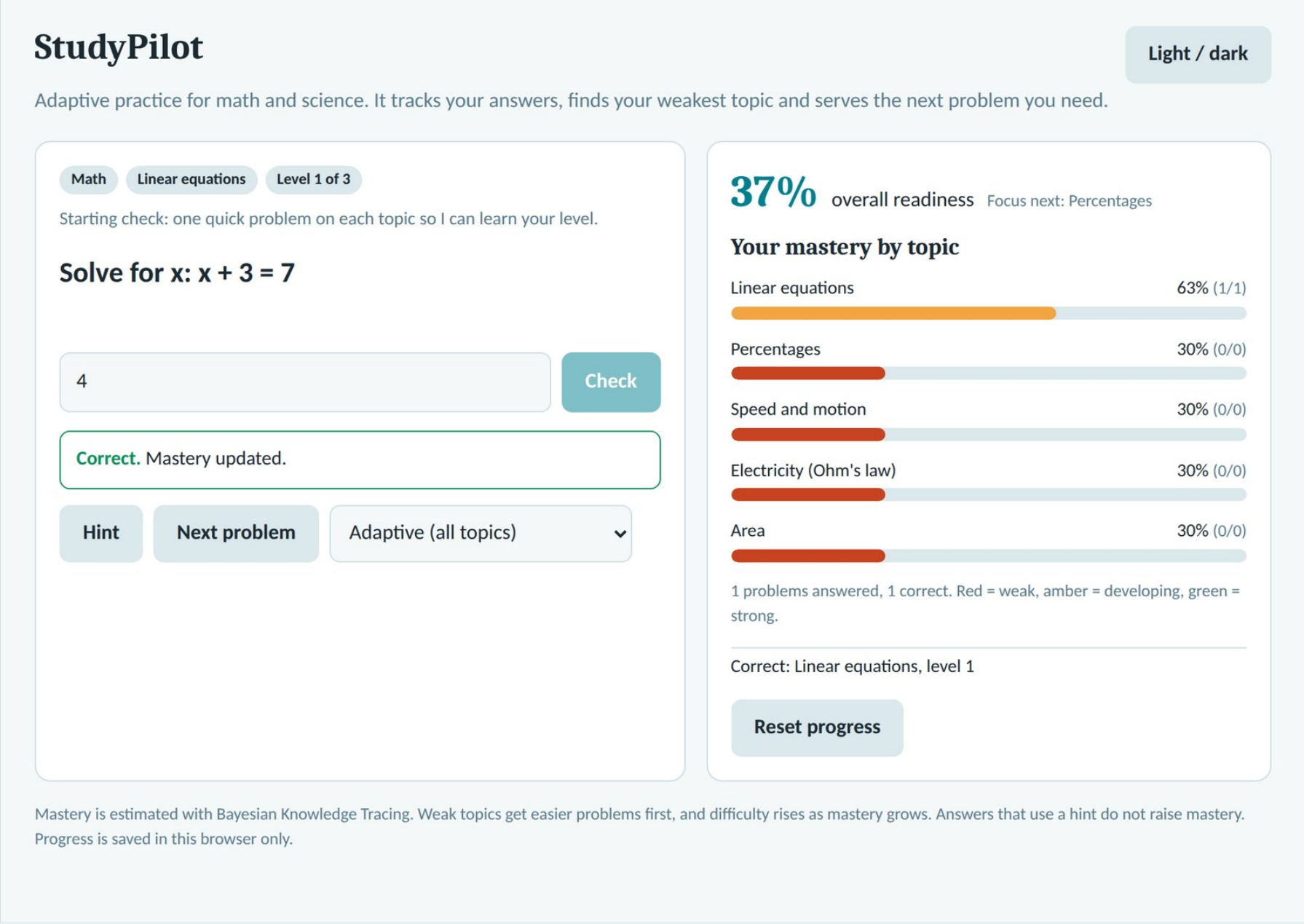
Screenshots were captured from the working prototype. The practice history in the dashboard screens comes from a sample session of 25 answers.

Home screen: first diagnostic problem



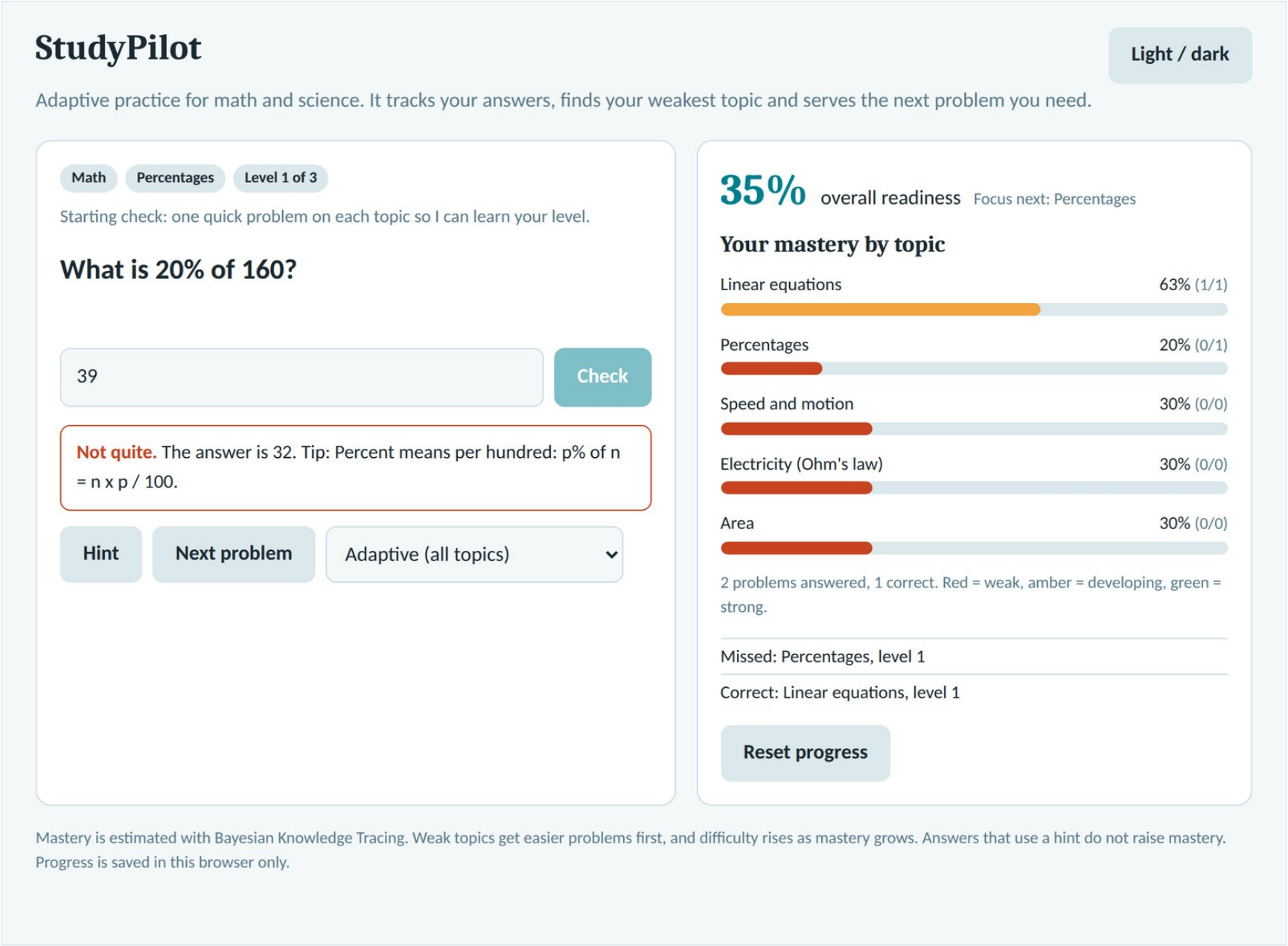
A new student starts with one quick level 1 problem per topic. The practice card on the left shows the subject, topic, difficulty level and why this problem was chosen. The dashboard on the right starts empty.

Instant feedback after a correct answer



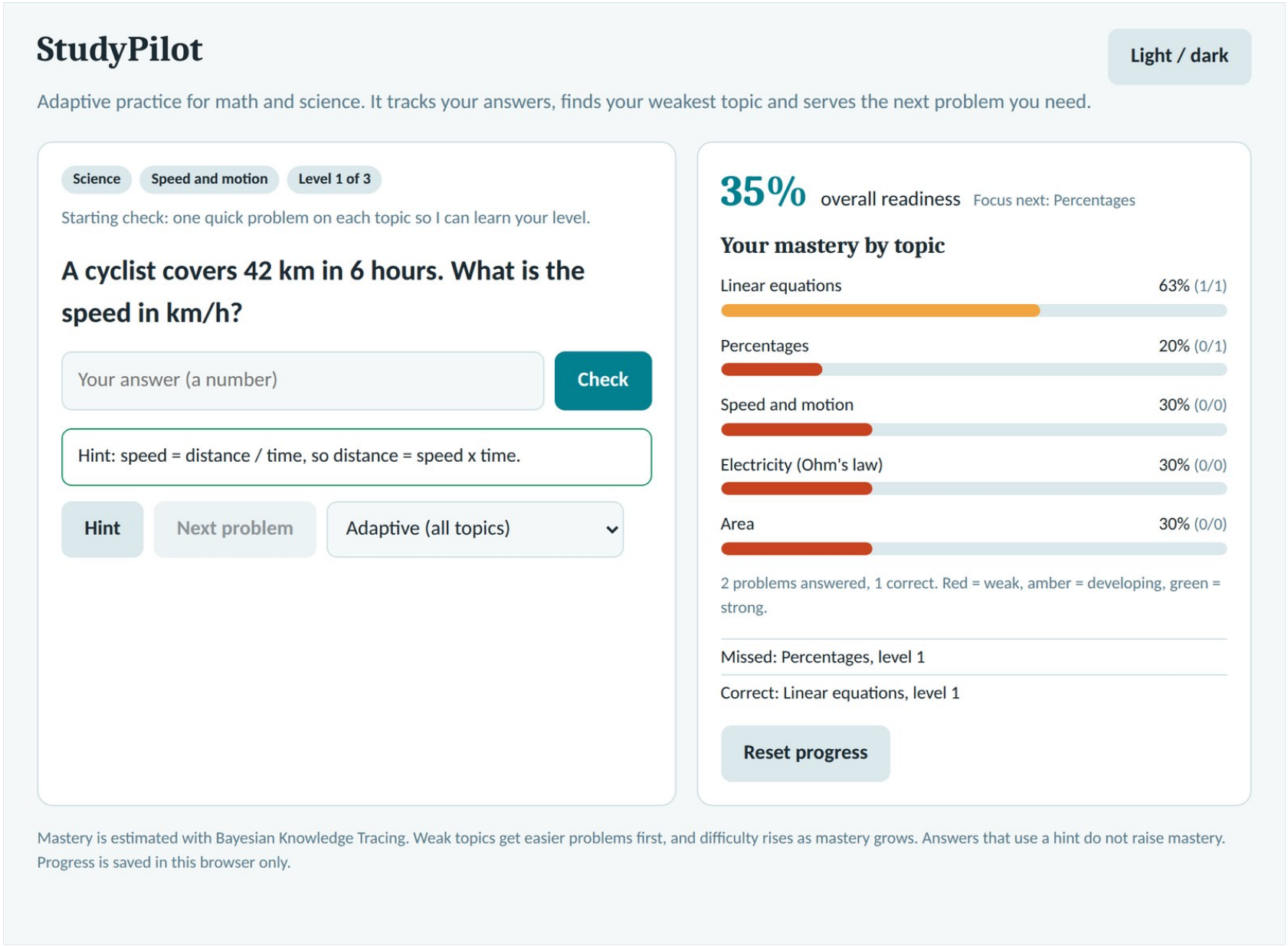
The student types a numeric answer and presses Check. The app confirms the result, updates the topic's mastery score and enables the Next problem button.

Feedback after a wrong answer



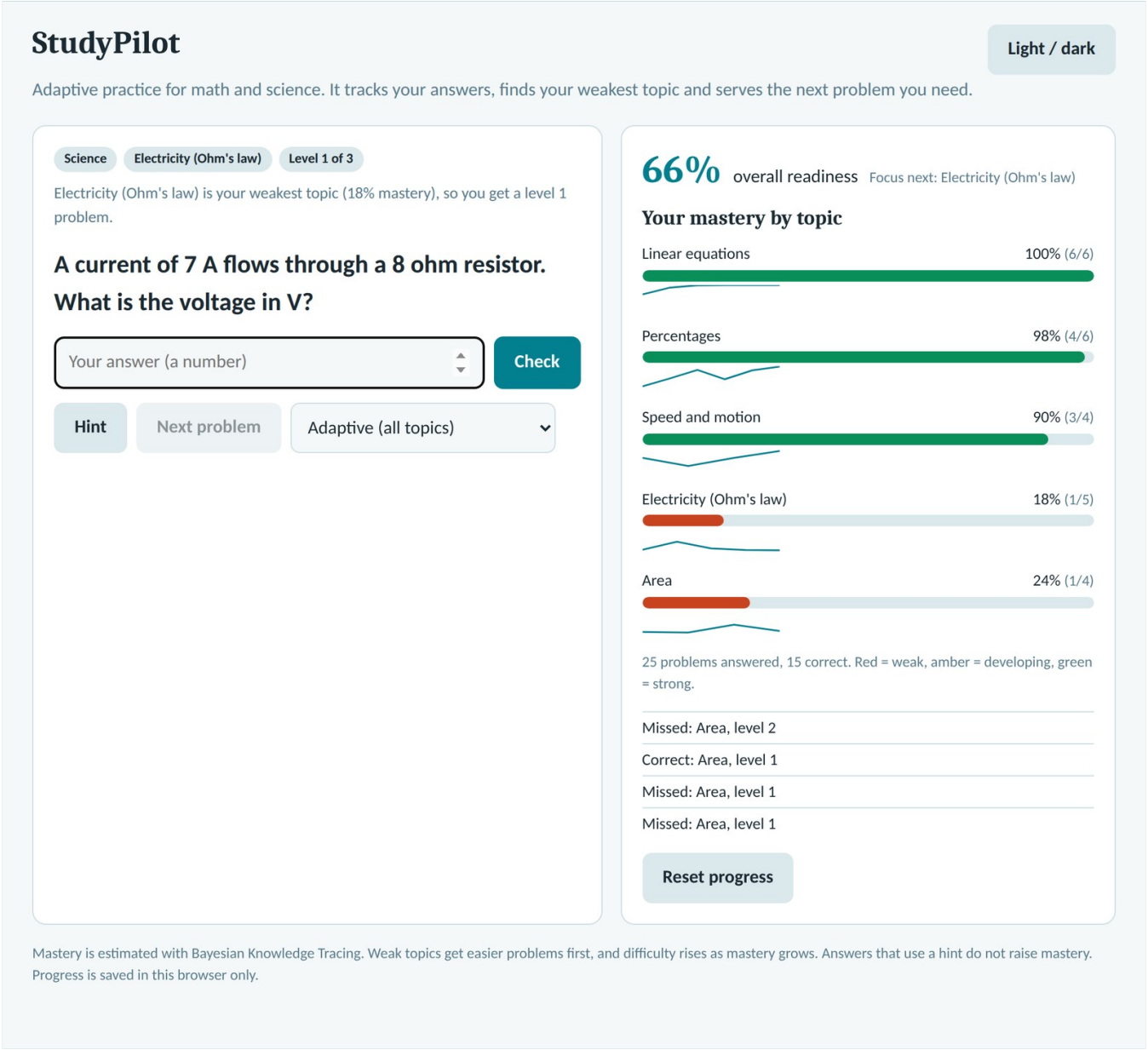
A wrong answer shows the correct result and a short tip for that topic, then lowers the mastery estimate for the topic.

Hint button



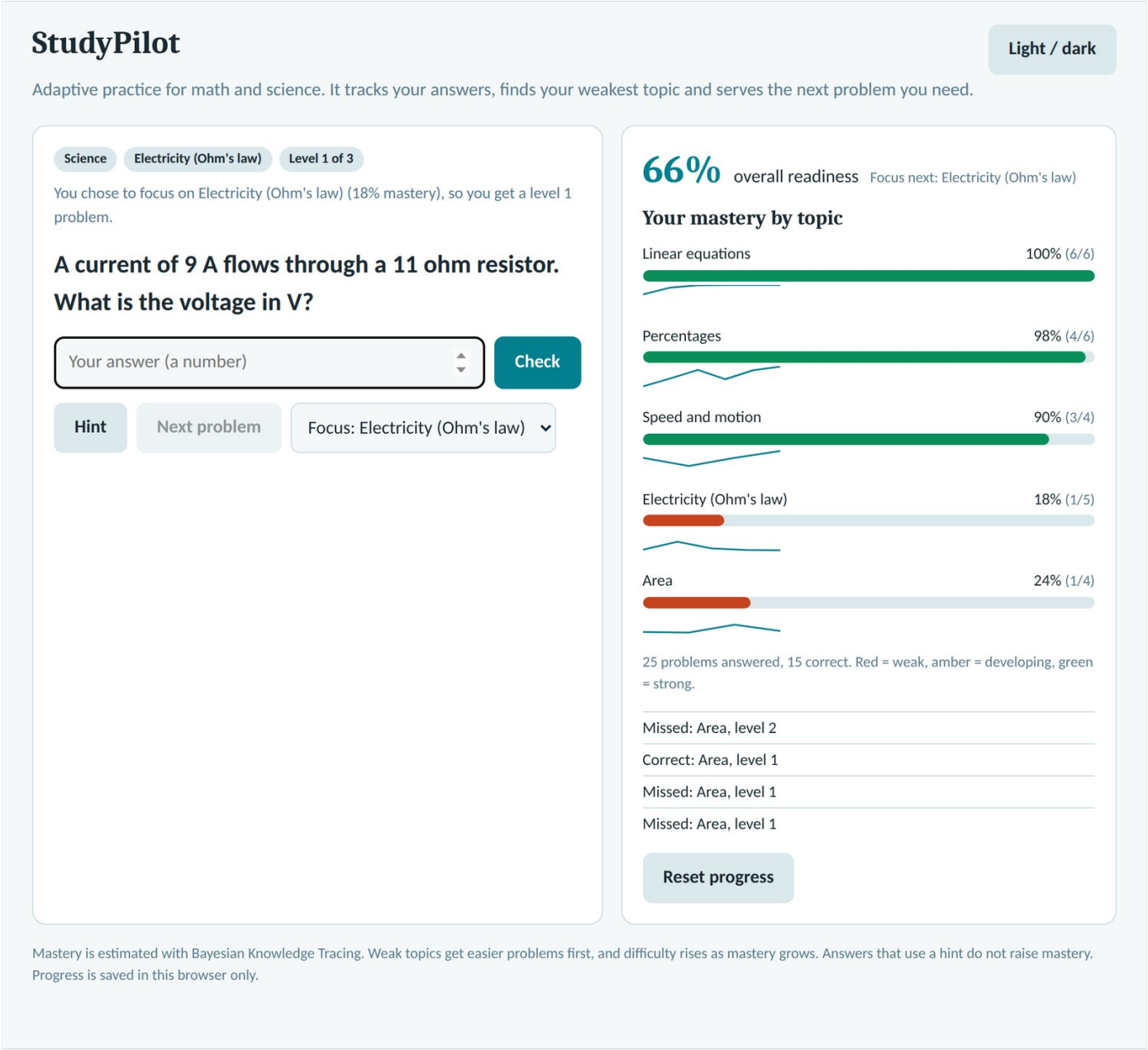
The student can ask for a hint before answering. A correct answer given after using a hint is recorded but does not raise mastery, so the score reflects unaided understanding.

Mastery dashboard after a practice session



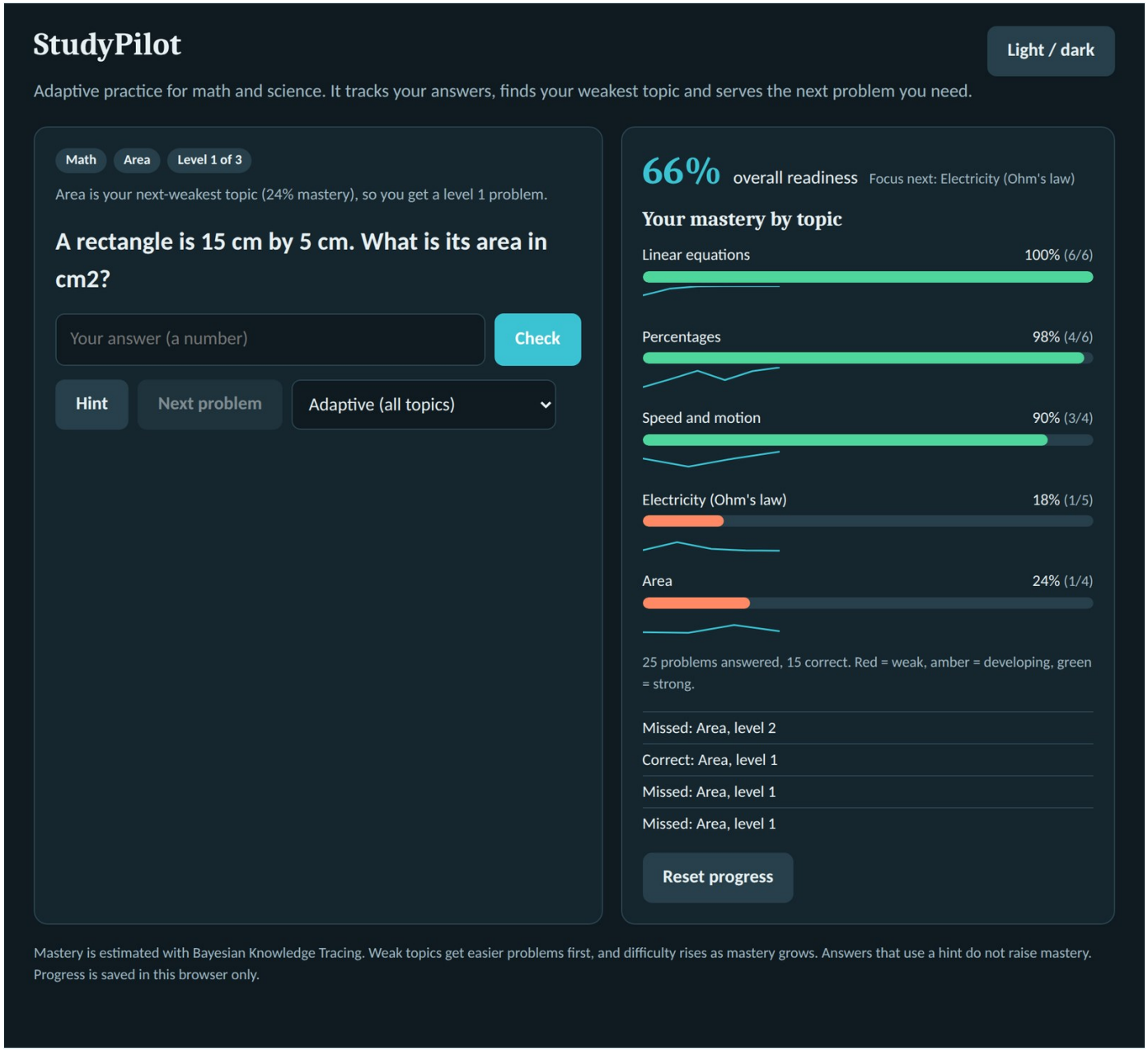
After 25 answers the dashboard shows an overall readiness score, a coloured mastery bar and trend line for each topic, and a recent-answers log. Red marks weak topics, green marks strong ones, and 'Focus next' names the weakest topic. The next problem is chosen from the weakest topics.

Practice focus mode



The student can leave the engine on Adaptive or choose a single topic to drill. In focus mode the difficulty still follows the student's mastery in that topic.

Dark mode



A toggle switches between light and dark themes, and the app also follows the device setting by default.

Mobile layout

On a phone-sized screen the practice card and dashboard stack into one column so the app works on the devices students use most.

